What Will Remain After the AI and Crypto Bubbles?

Aug 29, 2025 |**WILLIAM H. JANEWAY**

CAMBRIDGE – The supply chain for capital nowadays is being distorted by three phenomena. The first two are speculative bubbles, which could spill over from the financial sector to the real economy (following the pattern of similar episodes over the past 400 years). The third is an unprecedented shock from the state, which is asserting an unprecedented role within our already complex political economy.

The American innovation supply chain emerged during World War II and evolved over three generations. Cash flowed from federal institutions to research universities, where it funded scientific research and technological innovation. In turn, the universities served as transmission belts for the commercial economy, licensing their intellectual property to established firms and, especially, to venture-capital-backed startups. But since Donald Trump's return to the White House, this system, which underpinned US global leadership in science and technology for generations, has been disrupted.

Bubble Trouble

Let us first consider the bubbles. Financial markets – both public and private – are currently showing signs of two parallel speculative frenzies: one in cryptocurrency assets, and another in shares of AI-related companies. The crypto market is a bubble by definition, because there is no fundamental source of underlying value. Such assets do not generate cash flows for those who hold them; their current value depends upon expectation of greater resale value. A good parallel is the Dutch tulip mania during the 1630s, when the object of speculation similarly lacked any fundamental underpinning.

The AI hype cycle is a more common species of bubble. Once again, an innovative technology has emerged whose fundamental, long-term economic consequences cannot be known in advance. The bubble may have already peaked with this month's anti-climactic launch of OpenAI's GPT-5 model, but only time will tell.

A common feature of both bubbles is the extent to which investors have been prepared to pay super-premium valuations for securities with minimal liquidity and no governance rights. The flow of funds – from retail and institutional investors alike – into highly speculative illiquid assets has reached an unprecedented scale. Both bubbles emerged originally from an extraordinary financial environment: negative real interest rates and negative real returns on risk-free assets. Once they began to inflate, all the usual dynamics kicked in, driven by investors' fear of missing out.

The first law of financial bubbles is this: it is easy to know when you are in one, but difficult to know when it will pop. Still, those who study the issue have identified three signals that tend to mark the beginning of the end. The first is when the demand curve inverts, meaning that demand *increases* as prices rise. Two highly respected financial economists, José Scheinkman of Columbia University and Hyun Song Shin of the Bank for International Settlements, called attention to the occurrence of this phenomenon during the internet/dot-com bubble of the late 1990s and in the run up to the 2008 global financial crisis.

The second signal is when the exponential increase in price calls forth new supplies as many others try to get in on the action. Even in the digital world, it takes longer to generate a new asset than it does for the price to move. In the case of crypto, price moves are instantaneous; similarly, private equity markets move much faster than anyone hoping to build a new large language model (LLM) can. Finally, in the terminal stages of a bubble, demand is increasingly fed by uninformed, amateur investors.

Diverging Paths

All three signals appear to be flashing red in the crypto and AI markets. But the AI and crypto paths are likely to diverge, at least in terms of the catalysts that will burst the bubbles.

The price of crypto is of course dependent on demand, which can come from increased purchases by existing holders and from new buyers. Increased demand turns on the Trump administration's activist program of deregulation – an agenda that is impossible to separate from the administration's unprecedented corruption (through the issuance of presidential meme coins and the like). Thus, a continuation of crypto's run seems to depend on Trump and his cabal retaining political power. Given all the resources that the crypto industry has poured into lobbying and campaign finance, this political underpinning may well be secure through the 2026 midterm elections and even beyond.

The AI bubble is different. Sooner or later, today's high valuations will require support from the underlying fundamentals. That means generating positive cash flows from the massive investments being made in computing infrastructure (data centers and the like). Unlike crypto (and unlike Dutch tulip bulbs 400 years ago), those betting on AI need an economically sustainable business model.

True, some businesses and entrepreneurs have identified economically relevant and commercially profitable applications for LLMs. But for current AI valuations to be even remotely sustainable, two conditions must be met. First, AI applications must be sufficient in scale to generate validating cash flows on the massive funds invested. Second, and equally important, the economics must allow for a stable equilibrium, with positive cash flow for competing suppliers.

Meeting the second condition could be a problem. Given that the fixed costs required to offer AI services dominate the marginal cost of each unit of service, the economics are daunting. As and when price tends toward marginal cost, all players will lose money. That is why previous technological revolutions have often matured into either a stable oligopoly (which is difficult to maintain) or a regulated monopoly.

The histories of the railroads, electrification, and the internet are all relevant here. Each required massive investments in physical infrastructure before anyone had discovered viable, scalable applications for the new technology. Today, we take these industries for granted, forgetting that their evolution featured serial bankruptcies and a wide range of supportive state interventions to protect competitors from themselves. As so often happens, rational individual responses to incentives generated hugely destructive coordination failures. Through it all, massive speculative capital flows funded the construction of these transformational networks in a process punctuated by episodic financial crises.

In the case of AI, what analyst Nicolas Colin calls the "compute-energy stack" is the equivalent of the railroad tracks, the power plants and distribution grids, and the fiber-optic cables and server farms of the previous two centuries. Once again, capital must first flow into assets whose economic value cannot be known in advance. For the AI bubble to deliver long-term, economically stable, financially rewarding results without disruptive discontinuity would be unprecedented in the history of capitalism.

Killing the Golden Goose

During the twentieth century, the supply chain of capital to fund frontier science and technology underwent an institutional transformation. Until the 1980s, industrial research labs funded by the monopoly profits of the great technology-based monopolies – DuPont, AT&T, GE, IBM, Xerox – played a major role in the American innovation system. But in 1982, the Securities and Exchange Commission created an alternative application for monopoly rents when it ruled that stock buybacks do not constitute "market manipulation." Since then, this particular use of cash has only grown. In 2024, total buybacks by US public companies reached $942.5 billion, more than 50% higher than aggregate business investment in research and development.

Of course, by the 1980s, the federal government's mobilization of science to win World War II and the Cold War, as well as to launch a "War on Cancer," had matured, thereby offsetting the old tech monopolies' retreat. But that brings us to the third, most damaging development: the Trump administration's unprecedented assault on the American scientific research enterprise.

Upon returning to the White House, Trump wasted no time targeting strategic funding agencies such as the National Institutes of Health, and National Science Foundation, the Department of Energy's Advanced Research Projects Agency (ARPA-E), and the R&D Office of the Environmental Protection Agency. Worse, he is pairing these cuts with a frontal assault on the research universities that have long generated the scientific discoveries and technological breakthroughs that underpin American competitiveness. These institutions are now being crippled.

Orchestrated by the Office of Management and Budget director, Russell Vought, this program of destruction was publicly previewed in the far-right Heritage Foundation's notorious Project 2025 (which Vought helped author). While the ostensible purpose is to extirpate diversity, equity, and inclusion programs and any effort to address climate change, the deeper mission is to unwind the legacy of the New Deal and return the US to the political economy of the 1920s. Only the Department of Defense would be left more or less intact.

While it is impossible to quantify the long-term consequences of this program, there can be little doubt that it is bad news for the US economy and people everywhere. The American innovation economy has delivered material progress on a scale unseen in human history; but now it is being systematically crippled.

Whither VC?

The crypto and AI bubbles not only emerged before this onslaught; they also arose largely outside of the conventional VC cycle that Paul Gompers and Josh Lerner of Harvard Business School documented in what is now a canonical text of the industry. Although there have been some VC-backed crypto ventures, such as Coinbase, and some VC firms, such as Andreessen Horowitz, that aggressively established themselves as crypto advocates, most of the funding has come from retail investors.

Now, the potential for institutional sponsorship, thanks to deregulation, has given new life to retail speculation in crypto. Absent vigorous state sponsorship, it would already be obvious that this speculation has reached the self-destructive limit of any Ponzi scheme. Just look at the proliferation of "crypto treasury" companies, a model pioneered by MicroStrategy (recently re-named Strategy). Its crypto-cultist founder and executive chairman, Michael Saylor, seemed to have created a perpetual motion machine, financing the purchase of Bitcoin by repeatedly raising capital on the basis of a public market valuation that was a multiple of the value of the Bitcoin he acquired. Dozens of hopefuls have followed suit, but many are trading in public markets at a *discount* to their crypto holdings, with Strategy shares falling 15% in August. In other words, they have generated the second signal mentioned above (when inflated prices call forth new supply).

By contrast, funding for the real assets underlying the AI bubble has come mostly from the Big Tech platform companies. Some VCs have sought to participate in the AI bubble, even when doing so required suspending adherence to the VC principles of valuation and governance. During the first half of 2025, OpenAI raised no less than $40 billion from a mix of sources led by the notorious bubble investor SoftBank. In the second quarter, five quasi-VC deals raised more than $1 billion for AI-related companies.

Within the VC world, exits and distributions (realizing profits and allocating them to investors) have been minimal for four years running, and the industry itself has become bifurcated. On one side are firms that have determined to stay true to their inherited mission by restricting their size and seeking substantial governance participation in early-stage projects; on the other are those who have transformed themselves into private equity-scale asset gatherers and fee collectors. For the industry as a whole, investment increasingly focused on AI has drained the "dry powder" from the funds raised in the Unicorn years through 2021.

Even as the broader financial regime has normalized from the pre-2021 era – when unconventional monetary policies had created a hyper-speculative environment – the future architecture of VC remains uncertain. The issue is not just that the crypto and AI bubbles still need to run themselves out. More fundamentally, the industry can no longer count on the strategic supply chain of innovation orchestrated by the American state. Many of the links in that chain are now being weakened or broken by the Trump administration.

The VC model was born in the US, but its future may lie in China and Europe. The strategic role that it has long played on its home turf will diminish as the flow of science-based transformational technologies dries up. It took three generations to build that supply chain. Reconstructing it will not be easy.

WILLIAM H. JANEWAY

William H. Janeway is a distinguished affiliated professor in economics at the University of Cambridge and author of *Doing Capitalism in the Innovation Economy* (Cambridge University Press, 2018).

https://prosyn.org/SaHN115